

Lecture 33

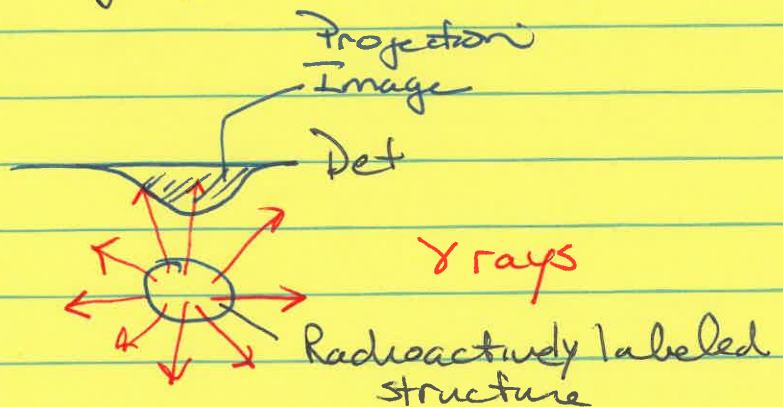
4/15/26
JRA

Projection Based Nuclear Imaging

Applications: Thyroid imaging
Bone cancers, Fractures
Lung & Kidney function

Want to get to the meat of the matter
This form of imaging, like x-rays, has two modalities

Projection



Suffers from

Take multiple views

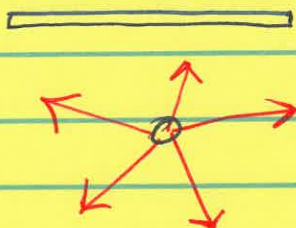
Important detector detail is the collimator

x-rays - use anti-scatter grid to reduce blurring due to scattering

RNI Use something similar (collimator)

What is collimator? why is it used?

Why
a collimator

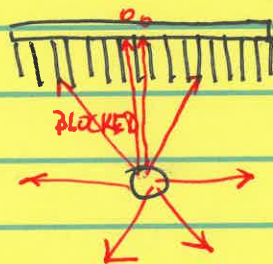


X-rays - emitted isotropically

Image is everywhere on detector because X-rays emitted in all directions

Doctor needs to address this to collect a "meaningful projection"

What
is collimator



Absorbing structures

Angled X-rays are blocked

Detect only X-rays that have some correlation w/ position

Detection depends on length of collimator

Resolution depends on length, among other factors, but noise increase because signal is lost.

Looked at PSF image
Determine width of PSF

Looked at similar triangle arguments

PSF in nuclear medicine $R_c = D \left(1 + \frac{X+c}{L} \right)$ Look up terms in equation

↑ Hole diameter ↑ Length of holes (diam of coll holes)

Want R_c to be small

To achieve good resolution

- * D small
- * L big
- * z small

Resolution is depth dependent
 z = object depth
(measured from end of collimator)
Better for small z

Doctor gets better resolution
for objects nearer surface

PET does NOT need a collimator

Strong signal

- D big
- L small
- z small due to attenuation

Antagonistic

Typical collimator resolution - see PP values
Write out when recopying notes

Doctor will do projection imaging

Nuclear Tomographic Imaging

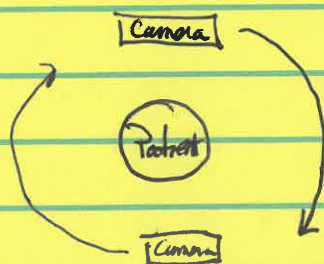
Two big techniques

- (1) SPECT: single photon emission computed tomography
- (2) PET: positron emission tomography

Projections - see advantages & disadvantages

These images can be smoothed
Main problem = increased dose

Nuclear CT take ≈ 128 100 views
This is about $1/10$ # taken w/ X-ray CT



Camera rotates around patient to take different views

Look at example w/ 4 sources

• •
• •

Take a lot of images

This technique uses single γ emitters

Popular source is $Tc-99m$

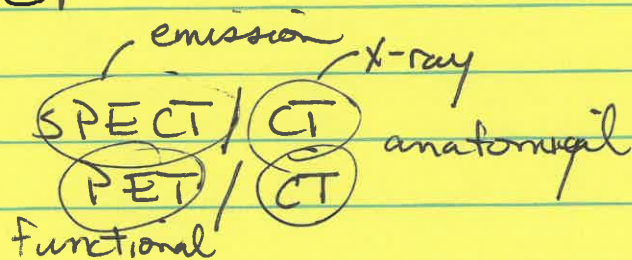
Like an excited fluorophore

Starts in excited state, emits w/ $T_{1/2} = 6$ hrs

Emits ~ 140 keV γ -ray

Can combine w/ CT

Most "scanners" are
dual mode



Usually superimpose two images

Reverse photos < First overview 1.2
Next SPEC

SPECT - cheaper but lower resolution
poorer image quality

Why? SPECT
uses collimator

PET No collimator better image quality
because no collimator

PET

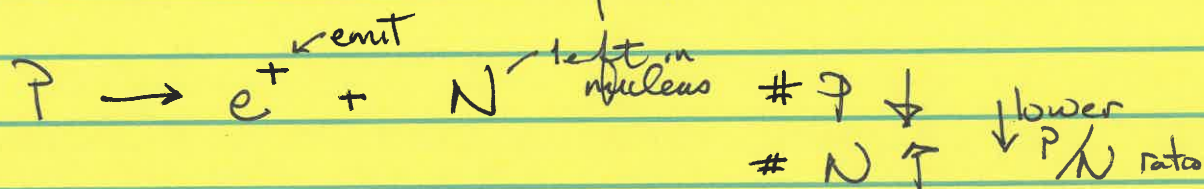
Uses positron emitter

Most popular = ^{18}F

unstable isotope
(prefers 19 electrons)

Positron emitter: excess of protons
 e^+ shortage of neutrons in nucleus

Proton in nucleus emits positron



Doctor cannot image e^+ too volatile

e^+ interacts w/ abundance of electrons in body

Positron quickly finds an electron
 e^+ e^-

e^+e^- Annihilate
when essentially
at rest

Then mass energy is converted
into γ energy

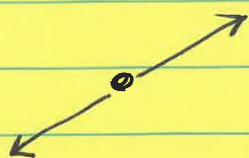
e^+ & e^- each have energy 511 keV (per $E=mc^2$)

↑ Very high energy
escape body $\rightarrow$ rays

$$\text{Momentum} = mv$$

" = 0 for annihilation

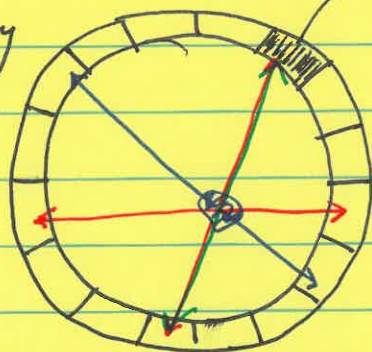
Momentum must equal zero after annihilation
 $\Rightarrow$ α -parallel γ -rays



This is really imp't
 $\Rightarrow$ No collimator required

Half life ≈ 2 hrs

Stationary
ring
detector



Individual detectors (hundreds)

Opposite detectors fire
 $\Rightarrow$ generate LOR
 line of response

Don't need collimator because know source is along LOR

Sample is at intersection of LORs

This is called "coincidence detection"

Fire $\approx$ simultaneously

PET is 2-3 orders of magnitude more sensitive than SPECT because not losing signal in collimator

Resolution in brain PET is $\approx$ few mm
This is good for nuclear imaging

Leave 2 things for Friday $\leftarrow$ Activity MRI

TOF Hot spot in center of ring $\Rightarrow$ 2 detectors will fire at same time
Hot spot away from center - measure one detector before another

Reg PET - prob uniform
TOF PET - get enhanced prob

Most PET devices, ^{now} TOF

Applications - looked at images